

Nicotinamide effect on the NAD⁺/NADH ratio in the diabetic rat retina

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Background: Diabetes mellitus is a chronic disease characterized by alterations in glucose metabolism that leads to complications in several organs, including the retina. Oxidative stress plays a key role in the progress of diabetic retinopathy and decrease in the NAD⁺/NADH ratio has been demonstrated in its pathogenesis. Therefore, NAD supplementation by nicotinamide (NAM) may be beneficial.

Methods: Four groups of Long-Evans rats were established: normal (control), normal supplemented with NAM, diabetic, and diabetic supplemented with nicotinamide (NAM). Diabetes was induced by a single intraperitoneal administration of streptozotocin (STZ). One day after STZ administration, rats received water containing 10 mM NAM for 5 h daily up to 20 or 45 days.

Results: Hyperglycemia induced increase in carbonylated proteins, as well as a progressive decrease in the NAD⁺/NADH ratio. Despite that, no significant changes in Nrf2 expression or GSH levels were found. NAM supplementation enhanced NAD⁺ levels in 20 days diabetic rat retina, reestablishing the NAD⁺/NADH ratio and carbonylated protein levels. However, NAM supplementation did not modify either NAD⁺/NADH ratio, nor levels of carbonylated protein at 45 days diabetes.

Conclusions: Alteration of the NAD⁺/NADH ratio induced protein oxidation independent of the Nrf2 function. Experiments aimed at clarifying the significance of these results are in progress.